Saathvik Maheshuni, sm13125@nyu.edu
Strategic Marketing (Fall 2026), Professor Jamyn Edis
November 3rd 2026

Garmin Venu Care: Extending Garmin into Monitored Fall Protection

1. The parent brand

To its owners, a Garmin is the watch you buy when you want the numbers right and the battery to last. One user put it bluntly: "The hardware is excellent, their software is poor" (Evans, 2026). Its points of parity are heart rate, GPS, notifications and payments (Garmin, 2026b). The one real point of difference is battery life. The 41 mm Venu 4 is rated up to 10 days, the 51 mm LTE fenix 8 Pro up to 27 days in smartwatch mode (Garmin, n.d.-b, 2025a). In a Reddit poll, 537 of 1,086 owners would leave Garmin or buy older models to keep low-power screens (Brecher, 2026).

Garmin's watches are a branded house, one name plus a series name (Garmin, 2026b). That lets the extension borrow Garmin's trust, app and Response center, but a failed safety service would hurt the whole Garmin name. Owners also resist paying monthly. In a TechRadar poll of 1,325 Garmin users, 80% never signed up for the $6.99-a-month Connect+ tier (Evans, 2026), so Venu Care's plan has to sell a real person picking up the call. Garmin's fiscal 2025 revenue was $7.25 billion, and the fitness segment, where Venu sits, had $2.36 billion, up 33% (Garmin, 2026b).

Garmin already has most of the pieces, but they were built for runners. Its LTE watches reach Garmin's own 24/7 response center by SOS, and the Forerunner 945 LTE sends it detected mid-run incidents (Garmin, 2025a, 2026a, 2026d, n.d.-a). The Venu 4's new fall detection only alerts contacts through a paired phone, not emergency services (Garmin, 2026e, 2026f).

2. The customer and the size of the prize

The segment is older adults who live alone, fell last year, and whose family worries. Most pick their own device (Clark, 2026). One in four adults 65 and older falls yearly (CDC, 2026).

The persona is made up, built from the survey numbers below. Marian, 79, lives alone, fell last spring and won't wear a pendant. Her daughter Karen wants to know she is OK; her worst fear is finding her mother after a night on the floor. Marian says she wants to be left alone, but really she wants to stay home, like 88% of older adults (University of Michigan poll, as cited in ACL, 2024). Her refusal is common: only 9% of adults 65 and older use a medical alert device, and 32% of non-users say they don't feel old enough for one (Clark, 2026).

TAM is 16.2 million adults 65 and older living alone (ACL, 2024), $12.47 billion at $769.87 each. It is from May 2024, the latest ACL profile. Cuts for a fall last year, an imperfect poverty screen (Pew Research Center, 2025) and current device users leave a SAM of 2.95 million wearers and $2.27 billion (Exhibit 1). I built SOM bottom-up from the budget and store shelf space. A $5.33 million media allocation, 2.3% of Garmin's 2025 advertising spend (Garmin, 2026b), at the benchmark cost per lead, converting one lead in four, buys 19,800 direct plans. One watch a week in 300 Best Buy stores, plus 600 on bestbuy.com, adds 16,200. That is 36,000 plans, 1.22% of SAM, $27.7 million at full price and $18.4 million to Garmin in Year 1.

3. The extension and where it sits

Garmin Venu Care is a 41 mm Venu-class watch (polymer case, glass lens, silicone band) with LTE-M, all-day fall detection, speaker, microphone and SOS button, sold only with the $34.99-

a-month Garmin Response Care plan. These specs are my assumptions. The seven-day battery target assumes care mode (fall sensor on, LTE standby, dim display), tested with up to two alerts a month, and launch depends on passing it. A fall the wearer does not cancel goes over LTE, without a phone, to a Garmin Response coordinator, who calls the wearer. If nobody answers, emergency services go first and family is told in parallel through the Caregiver app.

The Ansoff cell is diversification, a new product (it needs a monitored plan) for a new customer. Product development fits worse as the buyers are seniors and families buying monitoring, not sports-watch owners. Market development fails: the Venu 4 has no monitoring (Garmin, 2026e).

The frame of reference is monitored medical-alert watches. The two points of parity are 24/7 staffed two-way response with location and automatic fall detection that escalates. The point of difference is a week between charges. It matters because a watch on its charger is off the wrist. Only 18% of medical alert users wear theirs all the time, and 39% take it off to sleep (Clark, 2026). Garmin can plausibly deliver a week: its LTE Forerunner 945 is rated up to two weeks in watch mode (Garmin, n.d.-a), against incumbents' 6 to 24 hours. UnaliWear's Kanega stays on the wrist only by swapping batteries every 24 to 36 hours (UnaliWear, 2026). The reasons to believe are Garmin's battery record and the center it already runs (Garmin, 2025a, 2026a).

Exhibit 2 maps who answers a fall against days between charges. Exhibit 5 states the positioning: For older adults who live alone and have already fallen, and the adult children who worry about them, Garmin Venu Care is the monitored medical-alert watch that keeps automatic fall protection on the wrist a week at a time, because Garmin already builds watches rated for a week or more between charges and operates its own 24/7 Garmin Response center.

4. The price

The watch costs $202.60 to make and deliver (Exhibit 3), anchored to the Venu 4's realized price and the fitness segment's 37% cost of goods (Garmin, 2025b, 2026c). The service costs $18.15 a month per subscriber, mostly data and one eight-minute alert at the BLS telecommunicator wage (BLS, 2026; Hologram, n.d.). Three substitutes orient the price (checked October 29th 2026). LifeFone's Safe Watch Active is $635.40 in year one (AARP, 2026). Bay Alarm's SOS Smartwatch with Fall Detection costs $758.60, device plus twelve paid months (Bay Alarm Medical, n.d.-a). Apple's cellular Watch SE 3 is $299 (Abt Electronics, n.d.) with no monitoring fee, or about $419 with a carrier line. The ceiling is my guess. Every watch AARP reviewed charges $5 to $10 a month extra for fall detection (AARP, 2026), so one safety feature is priced at $60 to $120 a year. Bay Alarm's advertised $758.60 buys a watch rated 6 to 18 hours. Adding one feature's worth for a week of battery gives $819 to $879; I set $850 and priced below. Kanega, which stays on, charges $1,318.40, $560 over Bay Alarm (UnaliWear, 2026).

The price is $349.99 plus $34.99 a month. I used perceived-value pricing. The watch is priced above rivals' hardware, since it doubles as a daily smartwatch, and the plan below their $49.95 to $52.95. I am skimming on the device ($349.99 against rivals' $0 to $299) with no launch discount, and will lower it after the 300-store trial. I judge first buyers, right after a scare, to be fairly inelastic: non-owners rank cost only fourth (Clark, 2026), so a discount would only anchor a low price. $34.99 reads as under $35, and $349.99 stays under a $350 search filter.

5. The route to market and the economics

I plan a U.S. launch on October 1st 2027 through garmin.com and one retail account, Best Buy, 55% and 45% of units, every plan activating through Garmin. Best Buy takes a 30% margin

($244.99 wholesale) plus a $24 plan share per activation, counted once in CAC. It brings reach and demos, and Garmin gives up some control over how the service is explained. I assume a trial in 300 of its 926 Domestic stores (Best Buy, 2026b). These roles are my guesses. The decider is a Category Manager for Wearables; the approver is plausibly Patrick McGinnis, chief revenue officer, who runs merchandising and Best Buy Health (Best Buy, 2026a). Best Buy Health, which owns Lively and took a $118 million goodwill impairment in fiscal 2026 (Best Buy, 2026b), is a likely gatekeeper, since Lively Mobile2 is its own medical alert and Lively dropped its response service on Apple Watch in August 2024 (Lively, n.d.). The merchant needs to hear that Venu Care brings the wearables aisle a full-margin care product for families who refused a medical alert device, about $6,700 a year per store (Exhibit 3). I can't yet prove those buyers are new.

Year 1 brings $18.4 million of revenue, $7.0 million of contribution after card fees and a $1.8 million loss after $8.8 million of acquisition spend (Exhibit 4). The Week 3 LTV uses $34.99 ARPU, 48.1% margin and 2.5% monthly churn, 26% a year, inside the 24% to 36% reported for medical alert accounts in 2013 but under the 30% to 40% a 2017 article expects (Security Systems News, 2013; Giacalone, 2017). Plan LTV is $674 of margin ($1,400 of revenue); with $94 of device profit, it is $768 before CAC. Blended CAC per activation is $243.61, with $289 direct ($67.36 per lead at 25% activation plus $20 creative; WordStream, 2026). Margin LTV is 3.15 times CAC, clearing the 3-to-1 rule, but 2.39 on the HBS calculator at a 12% discount rate.

6. The case against it

The strongest case against is a $299 cellular Apple Watch SE 3, set up from an adult child's iPhone, which calls emergency services and messages family after a fall, with no monitoring fee (Apple, 2026a, 2026b). Garmin's center handled over 3,000 satellite SOS incidents in 2025, and its spare capacity is unknown (Garmin, 2026a); 36,000 subscribers would send about 214,000 alerts in Year 1, and the 48% plan margin assumes one a month. The incumbents also charge $0 to $299 up front against $349.99 here, and LifeFone lets a family cancel at any time (LifeFone, n.d.-a). Only 8% of users rank battery life a top factor (Clark, 2026). In answer, Apple's watch runs 18 hours and calls 911 with no monitoring center, so no person calls the wearer first to check. The monitored watches need charging or a battery swap every 6 to 36 hours. Users rank monitoring first but ask for "longer battery life to enable overnight wear" (Clark, 2026). This is for families who will pay more up front for a week between charges. It needs hiring, training and tested response at $8 an alert. At 2.0 alerts a month, LTV is still 1.84 times CAC. A trial above 2.0 alerts, a failed battery test or a staffed Apple tier would change my mind.

I assume 10% of buyers, 3,600, would otherwise have bought a Venu 4, which now has a fall alert (Exhibit 6). That loses $0.92 million of Venu 4 profit ($0.22 million net of what they pay as Venu Care customers, which Year 1 already counts), so Year 1 overall is -$2.69 million. Development, my estimate $9.85 million, pays back from the first year's buyers in Year 5, undiscounted (Exhibit 6). The dilution risk is a missed fall in the news denting Garmin's name. An illustrative 1% to 2% dent in Americas fitness revenue costs $7.7 to $15.4 million of gross profit, 1.0 to 2.1 times Year 1 gross profit. The kill criterion is 90-day paid retention for the October 1st to November 30th 2027 cohorts, read March 1st 2028. Below 90%, where LTV falls to 2.4 times CAC at one alert a month and 1.4 times at two alerts a month (Exhibit 4), Garmin closes new enrollments and stops further Best Buy orders, still serving subscribers. Expanding past the 300-store trial has its own bar: 3 times CAC on real churn and alerts (Exhibit 4).

Exhibit 1. TAM, SAM and SOM (United States, Year 1 = October 2027 to September 2028)

Year 1 price per customer is $349.99 + 12 x $34.99 = $769.87. Numbers are from my Excel sheet and rounded.

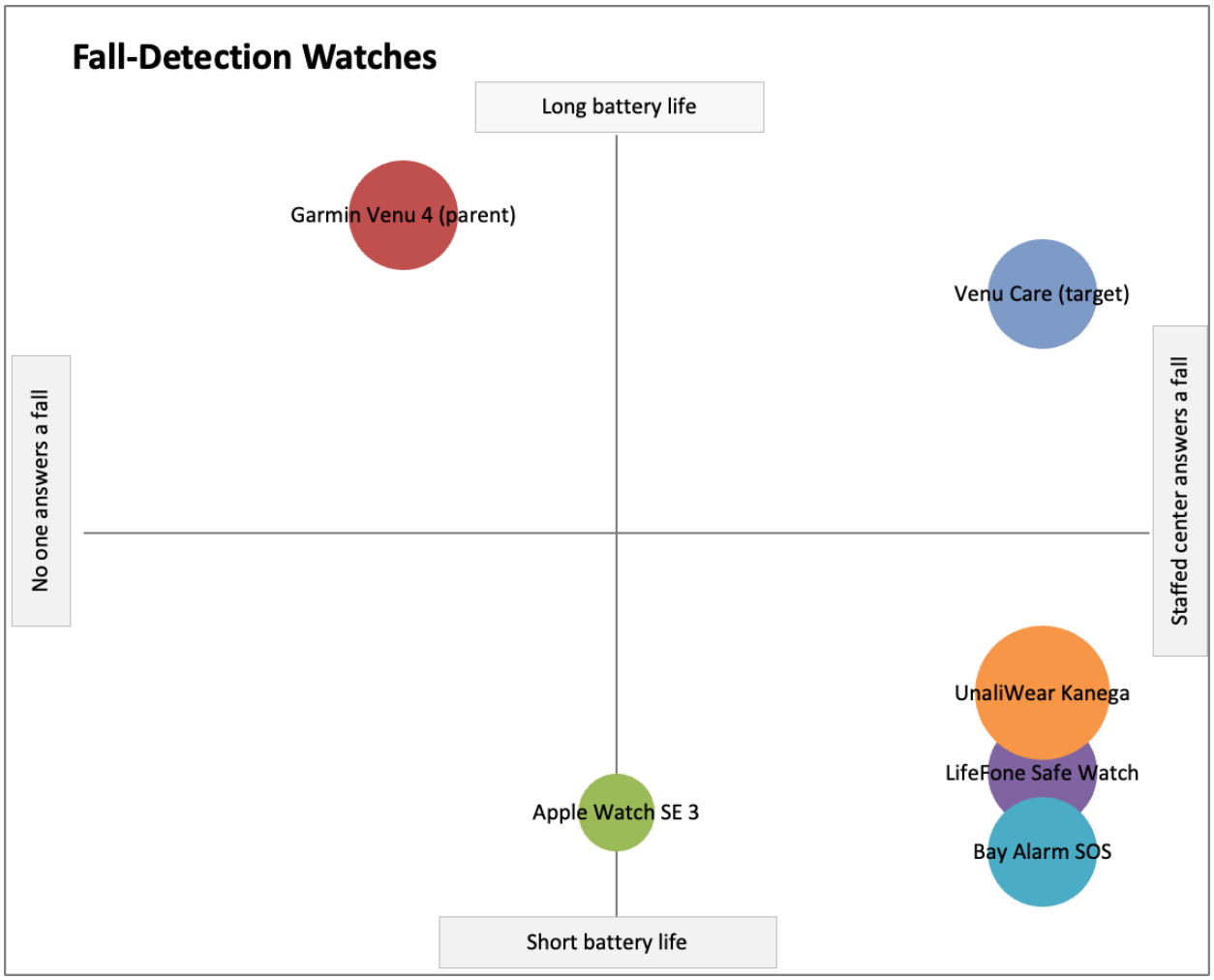
Layer	How I got it	People	Year 1 dollars
TAM	Adults 65+ living alone: 16.2 million (ACL, 2024, p. 6) x $769.87	16,200,000	$12.47B
SAM, cut 1	x 25% who fell in the past year (CDC, 2026, "1 in 4")	4,050,000	
SAM, cut 2	x 80% not in poverty (Pew Research Center, 2025)	3,240,000	
SAM, cut 3	x 91% who don't already use a medical alert device (Clark, 2026: 9% do)	2,948,400	$2.27B
SOM, direct	79,200 leads at $67.36 each (WordStream, 2026) = a $5.33M media budget (my estimate; 2.3% of Garmin's $228.5M 2025 ad spend, Garmin, 2026b), x 25% who sign up (my estimate; healthcare phone leads convert at 40%, Invoca, 2025)	19,800	
SOM, Best Buy	300 stores x 1 watch a week x 52 weeks (my estimate) + 600 on bestbuy.com	16,200	
SOM, total	19,800 + 16,200 = 36,000, or 1.22% of SAM; x $769.87 = $27.7M at full price	36,000	$27.7M at full price
Year 1 revenue to Garmin	Watches $10.90M (Exhibit 3) + plans $7.48M (213,840 paid plan-months, 36,000 x 5.94, x $34.99)		$18.38M

Assumptions, one per layer:

- TAM: one watch per person; family members who pay are not counted again. The ACL profile (May 2024) is the latest one.
- SAM: all three cuts apply to the same people. The CDC fall rate and the 9% device use are for all adults 65+, and poverty is only a rough test of who can afford it. I cut current users because the plan is aimed at people who refused a device; anyone who switches is extra. I did not cut for LTE-M coverage because it is checked at signup, so SAM is a bit generous.
- SOM: built from the media budget and Best Buy shelf space. Customers join evenly through the year.
- Plan-months (from my Excel sheet): people join evenly through the year and 2.5% cancel each month, so each month's group keeps 97.5% of its people the next month. Adding up each group's paid months, an October joiner pays about 10.5 months in Year 1 and a September joiner pays 1; the average over the 12 groups is 5.94, so 36,000 x 5.94 = 213,840 (about 214,000) plan-months. Doing the same for the next 12 months gives 8.92 paid months each in Year 2.

Source: ACL (2024); CDC (2026); Pew Research Center (2025); Clark (2026); WordStream (2026); Invoca (2025); Best Buy (2026b); Garmin (2026b); my calculations and estimates.

Exhibit 2. Perceptual map: fall-detection watches for someone who lives alone



This Perceptual Map shows each watch scored from 1 to 9. Across: who answers when the watch detects a fall (1 = no one, 3 = a text to family, 5 = an automatic 911 call, 9 = a staffed center that calls you, then sends help and tells family). I chose it because users rank monitoring first (Clark, 2026). Up: days between charges, from each company's rating (1 = half a day or less, 9 = ten days or more). I chose it because a watch on its charger is off the wrist, and only 18% of users wear theirs all the time (Clark, 2026). Placements are my judgment, and Venu Care's seven days is a target, so it is labeled as a target. Circle size is what a family pays in the first year: small under $500, medium $500 to $1,000, large over $1,000 (Exhibits 3A, 3C and 3D).

Watch	**Who answers a fall (score)**	**Days between charges (score)**	**Why I placed it there**
Garmin Venu 4 (parent)	3	10 (9)	Fall alert texts contacts through a paired phone (Garmin, 2026e); 41 mm rated up to 10 days (Garmin, n.d.-b)
Apple Watch SE 3	5	0.75 (1.5)	Calls 911 automatically, no monitoring center (Apple, 2026b); up to 18 hours (Apple, n.d.)

Watch	Who answers a fall (score)	Days between charges (score)	Why I placed it there
LifeFone Safe Watch Active	9	1 (2)	Staffed 24/7 center; up to 24 hours (LifeFone, n.d.-a)
Bay Alarm SOS Smartwatch	9	0.5 (1)	Staffed center; 6 to 18 hours (Bay Alarm Medical, n.d.-b)
UnaliWear Kanega	9	1.25 (3)	Staffed center; batteries swapped every 24 to 36 hours without taking it off (UnaliWear, 2026)
Garmin Venu Care (mine)	9	7, target (8)	Staffed Garmin Response; seven-day target in care mode, tested before launch

Source: Garmin (2026e, n.d.-b); Apple (2026b, n.d.); LifeFone (n.d.-a); Bay Alarm Medical (n.d.-b); UnaliWear (2026); Clark (2026); placements are mine; map drawn in the course's Perceptual Map worksheet (perceptualmaps.com).

Exhibit 3. Price build: floor, substitutes, ceiling and my price

A. Floor: cost to make and deliver one watch

Item	Cost	How I estimated it
Basic Venu-class watch	$151.60	Average price Garmin gets for a Venu 4, $409.74 (my estimate: 15% sold direct at $549.99, 85% through stores at $384.99, which is $549.99 less a 30% store margin; Garmin, 2025b) x 37%, Garmin's fitness cost of goods (Garmin, 2026c)
LTE-M modem and antenna	$22.00	My estimate: Apple charges $50 more for cellular ($299 vs $249; Abt Electronics, n.d.; Apple, n.d.) x 37% ≈ $18.50, plus about $3.50 for the antenna and testing
Bigger charging cradle, bigger box with a large-print guide, extra freight and import duty	$18.00	My estimates: $6 + $3 + $9
Swap and warranty reserve	$11.00	My estimate, about 3% of the $349.99 price
Total per watch	**$202.60**	Card fees on direct sales (3% x $349.99 = $10.50) are counted separately

B. Floor: monthly cost of the service, per customer

Item	Per month	How I estimated it
Cellular data and voice	$4.00	Hologram: $1 + 50 MB x $0.03 = $2.50 (Hologram, n.d.), plus $1.50 for voice (my estimate). Garmin's own LTE plan is $7.99 (Notebookcheck, 2025)
Response staff	$8.00	1 handled alert a month (my estimate) x 8 minutes x $0.96 a minute (BLS median $25.50 an hour (BLS, 2026) + 35% benefits, staff busy 60% of the time) = $7.65, rounded up to $8
Customer support	$2.00	One 8-minute call every four months: 8 x $0.96 / 4 = $1.92, rounded to $2
Card fees	$1.05	3% x $34.99

Item	Per month	How I estimated it
Desk overhead, caregiver app and liability reserve	$3.10	My estimates: $1.25 + $0.35 + $1.50 (no insurance quote)
Total per month	**$18.15**	Margin: $34.99 - $18.15 = $16.84 a month (48.1%)

C. Orienting point: what families pay today for the first year (prices checked October 29th 2026)

Watch	Device	Monthly	First year
LifeFone Safe Watch Active	$0	$47.95 + $5 fall detection = $52.95 (AARP, 2026; LifeFone, n.d.-a); yearly billing is $48.95 a month (LifeFone, n.d.-b)	12 x $52.95 = $635.40
Bay Alarm SOS Smartwatch with fall detection	$159.20 (20% off $199)	$49.95 (Bay Alarm Medical, n.d.-a)	$159.20 + 12 x $49.95 = $758.60
Apple Watch SE 3 cellular	$299 (Abt Electronics, n.d.)	No monitoring; about $10 for a phone line (my estimate)	$299 + 12 x $10 = about $419
For reference: UnaliWear Kanega	$299	$84.95, or $779.40 a year (UnaliWear, 2026)	$299 + 12 x $84.95 = $1,318.40 ($299 + $779.40 = $1,078.40 on the yearly plan)

D. Ceiling, my price and what each channel earns

Item	Value	Working
Ceiling (my hypothesis)	$850 for the first year	Bay Alarm's $758.60 + one safety feature at $5 to $10 a month = $60 to $120 a year (AARP, 2026), so $818.60 to $878.60; I used the middle, rounded to $850, still under Kanega's $1,318.40, which stays on the wrist and costs $1,318.40 - $758.60 = about $560 more than Bay Alarm (Part C)
My price	$349.99 + $34.99 a month	$349.99 + 12 x $34.99 = $769.87 for the first year; perceived-value pricing; skimming on the device price, lowered after the trial; no launch discount
Best Buy margin	$105.00 a watch	30% of $349.99; Garmin gets $244.99
Garmin's profit per watch	$147.39 direct, $42.39 through Best Buy	$349.99 - $202.60; $244.99 - $202.60
Average device profit after card fees	$94.37	(19,800 x $147.39 + 16,200 x $42.39 - $0.21M card fees) / 36,000
Best Buy's gain per store	about $6,700 a year	($105 + $24 plan share) x 52 weeks at one watch a week

Year 1 watch revenue: 19,800 x $349.99 + 16,200 x $244.99 = $10.90M.

Source: Garmin (2025b, 2026b, 2026c); Apple (2026a, 2026b, n.d.); Abt Electronics (n.d.); BLS (2026); Hologram (n.d.); Notebookcheck (2025); AARP (2026); LifeFone (n.d.-a, n.d.-b); Bay Alarm Medical (n.d.-a, n.d.-b); UnaliWear (2026); my estimates.

Exhibit 4. Lifetime value, cost to acquire, Year 1 money and what-if tests

A. Lifetime value, Week 3 form: LTV = (ARPU x margin) / churn, all monthly

Step	Working	Result
ARPU and margin	$34.99 a month; margin $16.84 (Exhibit 3)	48.1%
Monthly churn	0.52% deaths (National Center for Health Statistics [NCHS], 2026: 1.77%, 4.26% and 13.83% a year at 65-74, 75-84 and 85+, mixed 15/60/25 by me = 6.28% / 12) + 0.33% into care (4% a year / 12) + 1.65% cancelling (both my estimates)	2.5% a month
Churn check	2.5% a month = 26% a year, inside 24% to 36% for medical alert accounts (Security Systems News, 2013); a 2017 article says 30% to 40% (Giacalone, 2017); my kill line is about 34%	
Plan LTV	$16.84 / 0.025 (revenue LTV: $34.99 / 0.025 = $1,399.60)	$673.60
Plus device profit	Exhibit 3D	$94.37
LTV before acquisition cost	$673.60 + $94.37	**$767.97**
HBS LTV calculator (Basic Model)	Inputs: $34.99 a month x 48.13% margin = $16.84, 97.5% kept each month, 12% a year (0.9489% a month), 0% inflation. Calculator NPV = $16.84 / (0.009489 + 0.025) = $488.27, + $94.37 device profit	$582.64

Assumptions behind Part A:

- The price ($34.99) and churn (2.5%) stay the same every month.
- One handled alert a month, which sets the $8 staff cost (Exhibit 3B).
- Device profit counts once, at signup; no discounting except the HBS check at 12% a year.

B. Cost to acquire one paying customer (CAC)

Channel	Working	Cost
Direct	$67.36 per lead (WordStream, 2026) / 25% who sign up + $20 for ad creative (my estimate)	$289.44
Best Buy	$56 co-op + $24 plan share + $15 staff training (my estimates)	$95.00
Launch costs	$1.5M for agency, PR and store displays (my estimate) / 36,000	$41.67
Average CAC	(19,800 x $289.44 + 16,200 x $95 + $1.5M) / 36,000	**$243.61**
LTV / CAC	$767.97 / $243.61 (HBS: $582.64 / $243.61 = 2.39)	**3.15**

My bars: expand only at 3 times CAC; the trial must stay at or under 2 alerts a month (Part D).

C. Year 1 and Year 2 money (in $ millions, rounded, so totals can be off by 0.01)

Line	Working	Value
Revenue	$10.90 watches + $7.48 plans	$18.38
Gross profit before fees	$3.61 watches + 214,000 plan-months x $17.89 ($34.99 - $17.10 service cost before card fees) = $3.83	$7.43
Card fees	$0.21 on direct watches + $0.22 on plans	-$0.43
Contribution	$7.43 - $0.43	$7.00

Line	Working	Value
Acquisition spend	36,000 x $243.61 (Part B)	-$8.77
Year 1 result	$7.00 - $8.77	**-$1.77**
Year 2 margin from Year 1 customers	36,000 x 8.92 paid months each in Year 2 (worked like Exhibit 1) = 321,000 plan-months x $16.84	$5.41

D. What if I'm wrong? (CAC $243.61 unless noted)

Case	Working	LTV	LTV / CAC
Base	$673.60 + $94.37	$767.97	3.15
Kill line: 90% still paying at 90 days	(1 - c)^3 = 0.90, so c = 3.45%; $16.84 / 0.03451 + $94.37	$582.34	2.39
2 alerts a month (trial limit)	margin $8.84 / 0.025 + $94.37	$447.97	1.84
Kill line and 2 alerts	$8.84 / 0.03451 + $94.37	$350.52	1.44
Only 15% sign up direct	28,080 customers, CAC $306.68, device profit $82.37 (worked below)	$755.97	2.47

15% case: 79,200 leads x 15% = 11,880 direct + 16,200 Best Buy = 28,080 customers. CAC = (11,880 x ($67.36 / 0.15 + $20) + 16,200 x $95 + $1.5M) / 28,080 = $306.68. Device profit = (11,880 x $147.39 + 16,200 x $42.39 - 11,880 x $10.50 card fees) / 28,080 = $82.37.

To expand at 3 times CAC, churn must stay at or under 2.65% a month.

Source: WordStream (2026); NCHS (2026); Security Systems News (2013); Giacalone (2017); Harvard Business School Publishing (1999), LTV calculator in the course packet (Basic Model); my estimates.

Exhibit 5. Positioning statement

Positioning statement
For older adults who live alone and have already fallen, and the adult children who worry about them, Garmin Venu Care is the monitored medical-alert watch that keeps automatic fall protection on the wrist a week at a time, because Garmin already builds watches rated for a week or more between charges and operates its own 24/7 Garmin Response center.

Source: my statement; reason to believe from Garmin (2025a, 2026a, n.d.-b). The week between charges is a target that has to pass testing before launch (Section 3).

Exhibit 6. Cannibalization, dilution and development cost (the numbers behind Section 6)

A. Cannibalization: 10% of buyers would otherwise buy a Venu 4 (my estimate; I tested 5% to 15%).

- Venu 4 profit lost: 3,600 people x $255.67 ($409.74 - $151.60 - $2.47 card fee, which is 3% x $549.99 on the 15% sold direct; Exhibit 3A) = $0.92M
- Earned back: the same 3,600 bring $94.37 + 5.94 months x $16.84 = $194.40 each as Venu Care customers, or $0.70M
- Net Year 1 effect before CAC: -$0.22M (-$0.11M at 5%, -$0.33M at 15%)
- Year 1 overall: -$1.77M (Exhibit 4C, already counting these 3,600) - $0.92M = -$2.69M

B. Dilution, illustrative. I assume fitness follows Garmin's company-wide Americas share.

- Americas fitness sales: $1,998.0M in nine months x 4 / 3 = $2,664M a year, x 46% Americas share ($913.1M / $1,986.1M last quarter) = about $1,225M (Garmin, 2026c)
- A 1% to 2% dip: $12.2M to $24.5M of sales x 63% fitness gross margin (Garmin, 2026c) = $7.7M to $15.4M of gross profit, 1.0 to 2.1 times Venu Care's Year 1 gross profit of $7.43M (Exhibit 4C)
- Not costed: fitness buyers starting to see Venu as a medical brand. I would watch Venu 4 sales in the 300 trial stores against the other 626.

C. Development cost and break-even (my estimates, not in the Year 1 result)

Item	**Working**	**Cost**
Engineering	30 people for 18 months = 45 person-years x $173,266 (Garmin's 2025 R&D of $1,126.23M / about 6,500 engineering staff; Garmin, 2026b)	$7.80M
Training the response desk	46 coordinators at full size (36,000 x 8 minutes / 60 = 4,800 hours a month / 104 busy hours each) x 173 paid hours x $34.43 an hour (BLS wage + 35% benefits; BLS, 2026) x 2 months before launch	$0.55M
Certification and testing	My estimates: $1.00M certification and carrier approval + $0.50M battery and fall-alert tests	$1.50M
Total		**$9.85M**
Break-even	$9.85M / ($767.97 - $243.61 - $25.57 lost Venu 4 profit per customer, $0.92M / 36,000) = $9.85M / $498.79	about 19,750 customers, 55% of Year 1
Payback from Year 1 customers only	-$2.69M - $9.85M = -$12.54M; then +$5.41M (Exhibit 4C), +$3.99M, +$2.95M, +$2.17M in Years 2 to 5, each year keeping 0.975^12 = 74% of the one before	Positive in Year 5 (undiscounted)

Break-even counts each customer's whole lifetime; payback counts only the first year's buyers, year by year. On the HBS figure, break-even is about 31,400 customers (87% of Year 1). Discounted at 12% a year, Years 2 to 5 return $5.41/1.12 + $3.99/1.12^2 + $2.95/1.12^3 + $2.17/1.12^4 = about $11.5M, so payback slips past Year 5.

Source: Garmin (2025b, 2026b, 2026c); BLS (2026); my calculations and estimates.

References

AARP. (2026, August 25). *3 best medical alert watches of 2026: Tested and approved*. https://www.aarp.org/caregiving/home-care/best-medical-alert-watches/

Abt Electronics. (n.d.). *Apple Watch SE 3 GPS + Cellular 40mm Starlight aluminum case with S/M Sand sport band (MJLC4LW/A)* [Product page]. Retrieved October 29, 2026, from https://www.abt.com/Apple-Watch-SE-3-GPS-Cellular-40mm-Starlight-Aluminum-Case-with-S-M-Sand-Sport-Band-MJLC4LWA/p/248829.html

Administration for Community Living. (2024, May). *2023 profile of older Americans*. U.S. Department of Health and Human Services. https://acl.gov/sites/default/files/Profile%20of%20OA/ACL_ProfileOlderAmericans2023_508.pdf

Apple. (n.d.). *Buy Apple Watch SE 3* [Store page]. Retrieved October 29, 2026, from https://www.apple.com/shop/buy-watch/apple-watch-se

Apple. (2026a, October 27). *Set up Apple Watch for a family member*. https://support.apple.com/en-us/109036

Apple. (2026b, May 6). *Use Fall Detection with Apple Watch*. https://support.apple.com/en-us/108896

Bay Alarm Medical. (n.d.-a). *Medical alert costs / compare plans*. Retrieved October 29, 2026, from https://www.bayalarmmedical.com/pricing/

Bay Alarm Medical. (n.d.-b). *SOS Smartwatch: Medical alert watch*. Retrieved October 29, 2026, from https://www.bayalarmmedical.com/medical-alert-system/sos-smartwatch/

Best Buy. (2026a, July 14). *Best Buy announces Executive Team updates* [Press release]. https://corporate.bestbuy.com/2026/executive-team-updates/

Best Buy. (2026b, March 18). *Form 10-K for the fiscal year ended January 31, 2026*. U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/764478/000076447826000009/bby-20260131.htm

Brecher, H. (2026, April 23). *Garmin users want MiP instead of AMOLED smartwatches, survey shows*. Notebookcheck. https://www.notebookcheck.net/Garmin-users-want-MiP-instead-of-AMOLED-smartwatches-survey-shows.1281468.0.html

Centers for Disease Control and Prevention. (2026, September 4). *Older adult falls data*. https://www.cdc.gov/falls/data-research/index.html

Clark, A. (2026, June 22). *2026 medical alert device usage report*. The Senior List. https://www.theseniorlist.com/research/medical-alert-device-consumer-usage-study/

Evans, M. (2026, March 25). *'I'll switch to another brand': I polled over 1,000 Garmin users to see what they thought of Garmin's Connect+ premium tier, one year after its launch*. TechRadar. https://www.techradar.com/health-fitness/ill-switch-to-another-brand-i-polled-over-1-000-garmin-users-to-see-what-they-thought-of-garmins-connect-premium-tier-one-year-after-its-launch

Garmin. (n.d.-a). *Forerunner 945 LTE owner's manual*. Retrieved October 29, 2026, from https://www8.garmin.com/manuals/webhelp/GUID-2BA30752-33FB-42B8-8484-69C491706113/EN-US/Forerunner_945_LTE_OM_EN-US.pdf

Garmin. (n.d.-b). *Venu 4 series owner's manual: Battery information*. Retrieved October 29, 2026, from https://www8.garmin.com/manuals/webhelp/GUID-2CF5620C-E585-4E0A-9CC3-9565533EEE4D/EN-US/GUID-5B48351E-A637-46DE-A737-9A470C5D1B63.html

Garmin. (2025a, September 3). *Introducing fenix 8 Pro* [Press release]. https://www.garmin.com/en-US/newsroom/press-release/wearables-health/introducing-fenix-8-pro-the-first-ever-smartwatches-from-garmin-with-inreach-technology-for-satellite-and-cellular-connectivity/

Garmin. (2025b, September 17). *Take steps towards a healthier lifestyle with the Venu 4 from Garmin* [Press release]. https://www.garmin.com/en-US/newsroom/press-release/sports-fitness/take-steps-towards-a-healthier-lifestyle-with-the-venu-4-from-garmin/

Garmin. (2026a, February 25). *2025 inReach SOS year in review*. https://www.garmin.com/en-US/blog/outdoor/2025-inreach-sos-year-in-review/

Garmin. (2026b, February 18). *Form 10-K for the fiscal year ended December 27, 2025*. U.S. Securities and Exchange Commission. https://www.sec.gov/Archives/edgar/data/1121788/000119312526056028/grmn-20251227.htm

Garmin. (2026c, October 28). *Garmin announces third quarter 2026 results* [Press release]. https://investors.garmin.com/news-and-events/news/news-details/2026/Garmin-announces-third-quarter-2026-results/default.aspx

Garmin. (2026d, August 25). *Garmin expands its flagship performance smartwatch lineup with fenix 9 and fenix 9 Pro* [Press release]. https://www.garmin.com/en-US/newsroom/press-release/outdoor/garmin-expands-its-flagship-performance-smartwatch-lineup-with-fenix9-and-fenix9-pro/

Garmin. (2026e, September 22). *Garmin rolls out new feature updates for select smartwatches and cycling computers* [Press release]. https://www.garmin.com/en-US/newsroom/press-release/sports-fitness/garmin-rolls-out-new-feature-updates-for-select-smartwatches-and-cycling-computers/

Garmin. (2026f, September). *Venu 4 series owner's manual: Turning fall detection on and off* (Version 5). https://www8.garmin.com/manuals/webhelp/GUID-2CF5620C-E585-4E0A-9CC3-9565533EEE4D/EN-US/GUID-48CEA714-353F-4D14-BDFE-1C03B5846571.html

Giacalone, P. (2017, April 21). *Why now is the time to pursue PERS & mPERS*. Security Sales & Integration. https://www.securitysales.com/insights/why_pursue_pers_mpers/1351/

Harvard Business School Publishing. (1999). *Customer lifetime value calculator* [Course packet material].

Hologram. (n.d.). *IoT SIM card and global cellular connectivity: Pricing*. Retrieved October 29, 2026, from https://try.hologram.com/

Invoca. (2025, June 10). *Invoca releases definitive cross-channel and cross-industry buyer conversion benchmark report* [Press release]. CMSWire. https://www.cmswire.com/the-wire/invoca-releases-definitive-cross-channel-and-cross-industry-buyer-conversion-benchmark-report/

LifeFone. (n.d.-a). *At-home & on-the-go Safe Watch Active*. Retrieved October 29, 2026, from https://www.lifefone.com/at-home-and-on-the-go-safe-watch-active

LifeFone. (n.d.-b). *Order now / compare medical alert systems*. Retrieved October 29, 2026, from https://www.lifefone.com/order-now

Lively. (n.d.). *Lively app* [Help center page]. Retrieved October 29, 2026, from https://shop.lively.com/blogs/help-center/lively-app

National Center for Health Statistics. (2026, January 29). *Mortality in the United States, 2024* (NCHS Data Brief No. 548). https://www.cdc.gov/nchs/products/databriefs/db548.htm

Notebookcheck. (2025, September 3; updated 2026, February 18). *Garmin Fenix 8 Pro: New inReach smartwatch features explained*. https://www.notebookcheck.net/Garmin-Fenix-8-Pro-New-inReach-smartwatch-features-explained.1104361.0.html

Pew Research Center. (2025, December 4). *A smaller share of older U.S. adults live alone today than in 1990*. https://www.pewresearch.org/short-reads/2025/12/04/a-smaller-share-of-older-us-adults-live-alone-today-than-in-1990/

Security Systems News. (2013, September 25). *PERS company valuations predicted to rise*. https://www.securitysystemsnews.com/article/pers-company-valuations-predicted-rise

UnaliWear. (2026, June 23). *Frequently asked questions*. https://www.unaliwear.com/faq/

U.S. Bureau of Labor Statistics. (2026, October 29). *Public safety telecommunicators*. Occupational Outlook Handbook (wage data May 2025). https://www.bls.gov/ooh/office-and-administrative-support/police-fire-and-ambulance-dispatchers.htm

WordStream. (2026). *Google Ads benchmarks 2026* (data April 2025 to March 2026). https://www.wordstream.com/blog/2026-google-ads-benchmarks

AI Disclosure: I used Claude to help find relevant sources, double-check calculations, question some of my assumptions, and improve the clarity of my writing. I reviewed the information for accuracy and used it to structure the paper.